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marks from any two groups of mark points to max 6:

- 1 mutant TYR gene codes for non-functioning tyrosinase ;
- 2 albinism / albino, recessive (disorder / allele) ;
- 3 lack of melanin / no melanin produced ;
- 4 giving, pale / fair, skin / hair **or** giving, transparent / pink, iris ;
- 5 mutant HBB gene codes for, abnormal ( $\beta$ ) globin / haemoglobin S ;
- 6 sickle cell, anaemia / disease, codominant (disorder / allele) ;
- 7 less soluble / sticky / rod-forming, haemoglobin makes red blood cells sickle ;
- 8 sickle cells, block capillaries / cause pain (in low O<sub>2</sub> conditions) ;
- 9 mutant F8 gene codes for non-functional factor VIII ;
- 10 haemophilia, recessive / sex-linked / on X chromosome ;
- 11 fibrinogen not changed to fibrin ;
- 12 blood clotting problem / risk of excessive bleeding ;
- 13 mutant HTT gene codes for abnormal huntingtin protein ;
- 14 Huntington's disease dominant (disorder / allele) ;
- 15 *ref. to* additional / number of, CAG repeats ;
- 16 cognitive / movement, difficulty ;

*other human genetic conditions (max 2 conditions with 4 possible marks each)*

- 17 mutant named gene codes for abnormal named protein ; ;
- 18 type of allele that causes named condition ; ;
- 19 biochemical detail re. gene / protein / molecule ; ;
- 20 physiological effect ; ;

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